Interpreting the Solution of a System on a Graph

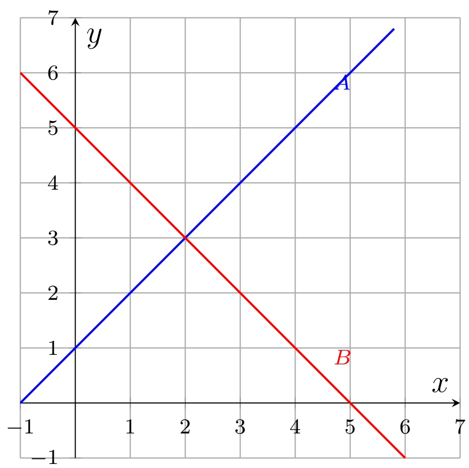
Reading a solution off a graph, telling one solution from none or infinitely many, and graphing to solve

Directions

- A point is a solution of a system only if it lies on *both* lines at once.
- Where a grid is already drawn on, read the point where the lines cross and write it as an ordered pair (x, y). Where the grid is empty, graph both lines first.
- Two lines can cross once, never (parallel), or everywhere (the same line). Say which.

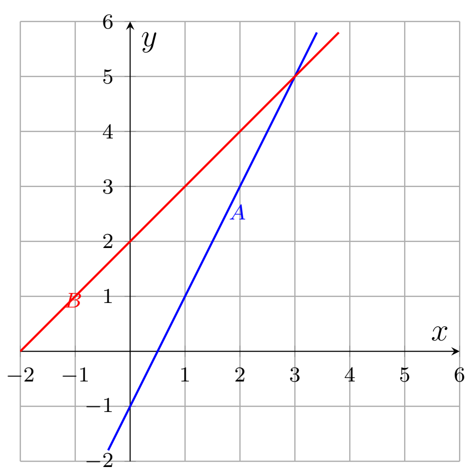
---

1. Line A is $y = x + 1$ and line B is $y = -x + 5$, both drawn on the grid. Write the solution of this system as an ordered pair.



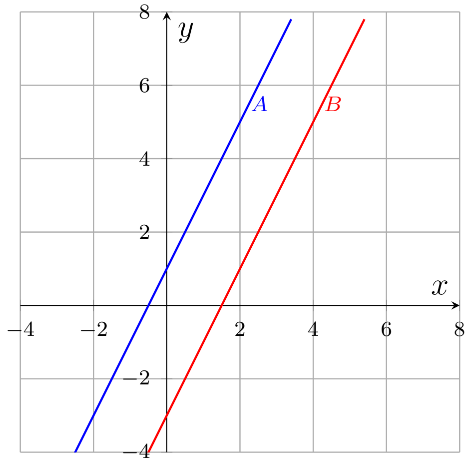
Answer: ____________________

2. Line A is $y = 2x - 1$ and line B is $y = x + 2$. Write the solution of this system as an ordered pair.



Answer: ____________________

3. The graph shows line A: $y = 2x + 1$ and line B: $y = 2x - 3$.



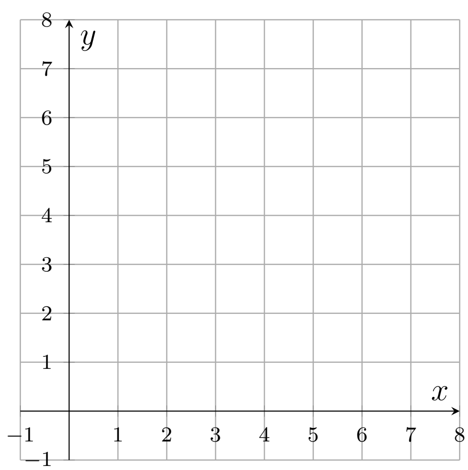
(a) How many solutions does this system have?

Answer: ________________

(b) Explain what the graph and the two equations tell you about why that is.

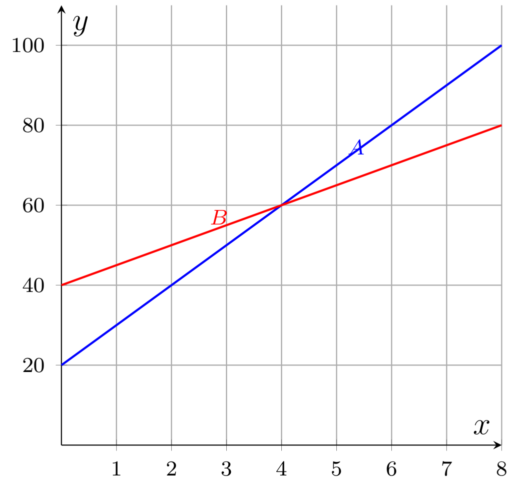
Answer: ________________

4. Graph $y = -x + 6$ and $2y = x$ on the empty grid, then write the solution of the system as an ordered pair.



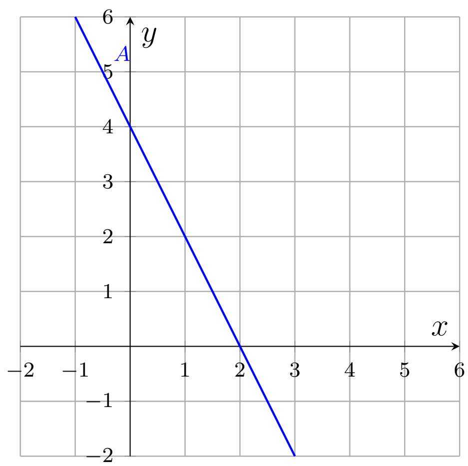
Answer: ________________

5. Two gym memberships are graphed: plan A costs $y = 10x + 20$ dollars after x months, plan B costs $y = 5x + 40$ dollars. Write the point at which the two plans have cost the same, as an ordered pair (months, dollars).



Answer: ____________________

6. The system is $y = -2x + 4$ and $4x + 2y = 8$. Only one line appears on the graph.



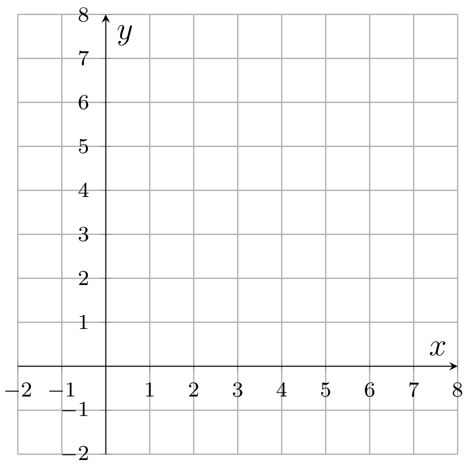
(a) Solve the second equation, $4x + 2y = 8$, for y.

Answer: ____________________

(b) How many solutions does the system have? Explain how part (a) accounts for what the graph shows.

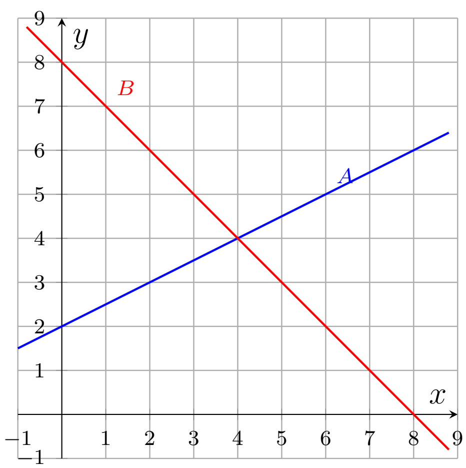
Answer: ____________________

7. Graph $2x + y = 8$ and $y = x - 1$ on the empty grid, then write the solution of the system as an ordered pair.



Answer: ________________

8. Line A is $2y = x + 4$ and line B is $y = -x + 8$.



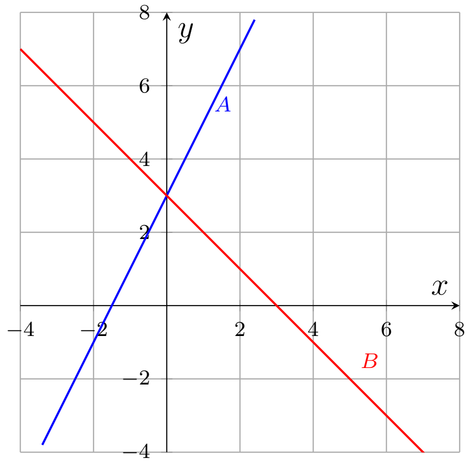
(a) Write the solution of the system as an ordered pair.

Answer: ________________

(b) The point $(6, 5)$ lies on line A. Explain why it is still not a solution of the system.

Answer: ________________

9. Line A is $y = 2x + 3$ and line B is $y = -x + 3$.



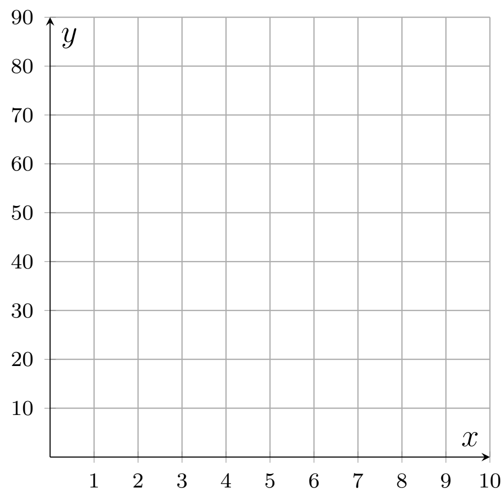
(a) Write the solution of the system as an ordered pair.

Answer: ______________

(b) Explain how the two equations let you predict that crossing point before graphing anything.

Answer: ______________

10. A van rental firm offers plan A at $y = 4x + 30$ dollars for x days and plan B at $y = 8x + 10$ dollars.



(a) Graph both plans on the empty grid.

(b) Write the break-even point as an ordered pair (days, dollars).

Answer: ______________

(c) Explain which plan costs less for rentals longer than the break-even point, and how the graph shows it.

Answer: ______________